

3.4 Projectiles

Mathematics B (MEI) — OCR B — A-Level (H640) — Mechanics — spec refs y1–y4

What this topic covers. Motion under gravity in a vertical plane — the modelling assumptions, the component equations, maximum height, range and time of flight, finding the initial velocity, and the Cartesian equation of the trajectory.

1. The modelling assumptions

The specification states these explicitly (spec ref y1):

- There is **no air resistance**.
- The projectile is a **particle**.
- The horizontal distance travelled is small enough that gravity always acts in the **same direction**.
- The vertical distance travelled is small enough that gravity is **constant** in magnitude.

Calculations involving air resistance are **excluded** from this specification. You may be asked to comment on the effect of introducing it — generally reducing both range and maximum height, and destroying the symmetry of the path — but not to compute with it.

2. The central idea

Horizontal and vertical motion are **independent**. Horizontally there is no acceleration, so the velocity is constant. Vertically the acceleration is g downwards, so the constant acceleration formulae apply. Time is the only quantity shared between them.

For a projectile launched at speed u at angle θ to the horizontal:

	Horizontal	Vertical (upwards positive)
Initial velocity	$u \cos \theta$	$u \sin \theta$
Acceleration	0	$-g$
Velocity at time t	$u \cos \theta$	$u \sin \theta - gt$
Displacement at time t	$u \cos \theta t$	$u \sin \theta t - \frac{1}{2}gt^2$

Resolve the initial velocity into components **before** doing anything else, and set out horizontal and vertical working in separate columns. Most errors in this topic come from mixing the two.

3. Key results

For projection and landing at the **same level**:

$$\text{time of flight } T = \frac{2u \sin \theta}{g}$$

$$\text{maximum height } H = \frac{u^2 \sin^2 \theta}{2g}$$

$$\text{range } R = \frac{u^2 \sin 2\theta}{g}$$

The maximum height occurs when the vertical component of velocity is zero — at exactly half the time of flight, by symmetry.

Worked example. A ball is projected at 20 m s^{-1} at 30° from level ground, with $g = 9.8$.
Components: $u \cos 30^\circ = 17.3 \text{ m s}^{-1}$ and $u \sin 30^\circ = 10 \text{ m s}^{-1}$.

$$\text{time to highest point} = \frac{10}{9.8} = 1.02 \text{ s}$$

$$H = \frac{10^2}{2(9.8)} = 5.10 \text{ m}$$

$$T = 2(1.02) = 2.04 \text{ s}, \quad R = \frac{u^2 \sin 2\theta}{g} = \frac{400 \sin 60^\circ}{9.8} = 35.3 \text{ m}$$

Work from the **exact** expression wherever possible. Multiplying the rounded values 17.3×2.04 gives 35.3 as well here, but compounding rounded intermediates through a longer calculation drifts the final figure — carry full accuracy and round once, at the end.

Maximum range

Since $R = \frac{u^2 \sin 2\theta}{g}$ and $\sin 2\theta$ is greatest when $2\theta = 90^\circ$, the range is maximised at $\theta = 45^\circ$, giving $R_{\max} = \frac{u^2}{g}$.

Because $\sin 2\theta = \sin(180^\circ - 2\theta)$, two different angles give the **same** range — θ and $90^\circ - \theta$. If 25° achieves a range, so does 65° . Questions frequently exploit this.

4. Horizontal projection

A body projected horizontally has $u \sin \theta = 0$, so its initial vertical velocity is zero and it falls exactly as if dropped.

Worked example. A ball is thrown horizontally at 15 m s^{-1} from a height of 20 m.

Vertically, $20 = \frac{1}{2}(9.8)t^2$, so $t^2 = 4.082$ and $t = 2.02 \text{ s}$.

Horizontally, $x = 15 \times 2.02 = 30.3 \text{ m}$.

A ball thrown horizontally and one simply dropped from the same height hit the ground at the **same time**. Horizontal motion has no effect on the vertical fall — a favourite examiner point.

5. Projection from a height

When the landing point is below the launch point, the flight is longer than the level-ground formula predicts. Taking upwards as positive and the launch as the origin, landing at depth h means the vertical displacement is $-h$:

$$-h = u \sin \theta t - \frac{1}{2}gt^2$$

Solve this quadratic for t and take the positive root.

The negative sign on h is the whole difficulty. Being consistent about the positive direction — and stating it — prevents the sign error that otherwise wrecks the answer.

6. Velocity and speed during flight

At time t the components are $u \cos \theta$ horizontally and $u \sin \theta - gt$ vertically, so

$$\text{speed} = \sqrt{(u \cos \theta)^2 + (u \sin \theta - gt)^2}$$

and the direction of motion is found from the ratio of the components.

At the highest point the vertical component is zero, so the velocity is **horizontal** and the speed is $u \cos \theta$ — not zero.

For level ground, the speed on landing equals the speed of projection, and the angle below the horizontal equals the angle of projection above it.

7. The Cartesian equation of the trajectory

Eliminating t between the components (y4):

$$x = u \cos \theta t \implies t = \frac{x}{u \cos \theta}$$

Substituting into the vertical equation:

$$y = x \tan \theta - \frac{gx^2}{2u^2 \cos^2 \theta}$$

This is a quadratic in x with a negative coefficient of x^2 , confirming that the path is a **parabola**.

Worked example. A projectile must clear a wall 3 m high at a horizontal distance of 20 m. What must be checked?

Substitute $x = 20$ into the trajectory equation and confirm that the resulting y exceeds 3.

Checking that the *range* exceeds 20 m, or that the *maximum height* exceeds 3 m, is not sufficient — the projectile might be too low at that particular horizontal distance. It is the height **at** $x = 20$ that matters.

8. Finding the initial velocity

Given enough information — typically the range and angle, or two points on the path — the launch speed can be recovered (y3).

Worked example. A projectile launched at 30° has range 50 m on level ground, with $g = 9.8$.

$$50 = \frac{u^2 \sin 60^\circ}{9.8} \implies u^2 = \frac{50 \times 9.8}{0.866} \implies u = 23.8 \text{ m s}^{-1} \text{ (3 s.f.)}$$

9. Two heights, two times

A projectile passes any given height (below its maximum) **twice** — once rising, once falling. The vertical displacement equation is a quadratic in t , and its two roots are exactly those two instants. Both are valid answers unless the question restricts to one.

10. Common pitfalls

- Using u rather than its components in the vertical equations.
- Applying gravity to the horizontal motion.
- Taking the speed at the highest point to be zero — only the *vertical component* vanishes.
- Sign errors when projecting from a height above the landing point.

- Giving only one root when a projectile passes a height twice.
- Forgetting the second angle that produces the same range.
- Checking the range or maximum height instead of the height at the obstacle's distance.
- Working in degrees or radians inconsistently between resolving and calculating.